



COMPUTER Network

22CSE62 –M.4.2



Module 4 -Transport layer and congestion control techniques

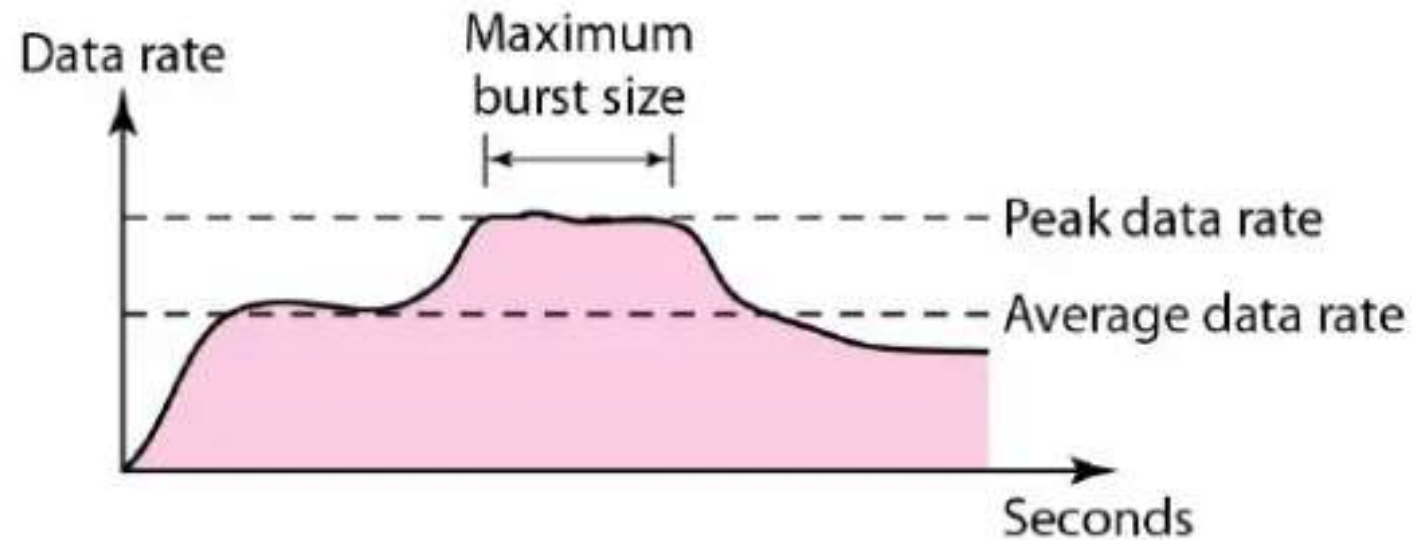


- Transport Layer: UDP , TCP , Data Traffic , Congestion , Congestion Control algorithms, Quality of Service, Techniques to improve QoS



Data Traffic

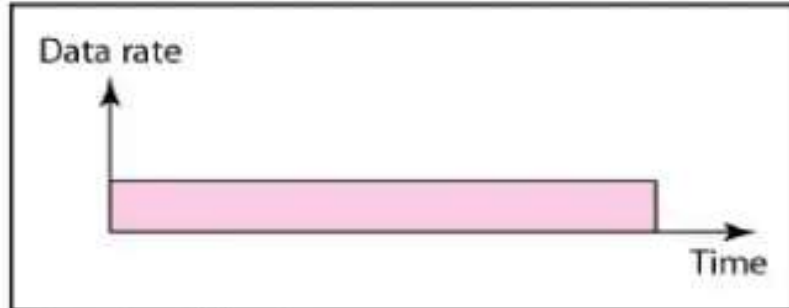
- The main focus of congestion control and quality of service is **data traffic**.
- In congestion control we **try to avoid** traffic congestion.
- In quality of service, we try to create an appropriate environment for the traffic.



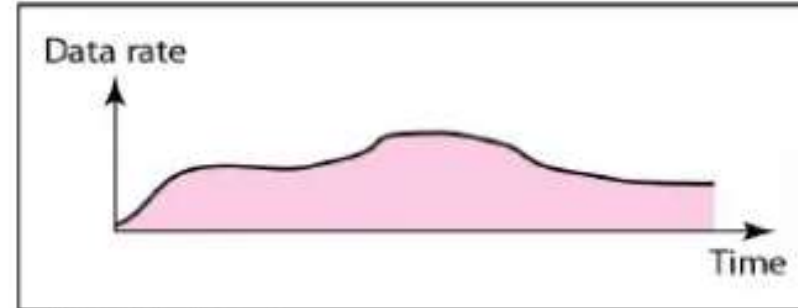
Traffic descriptors



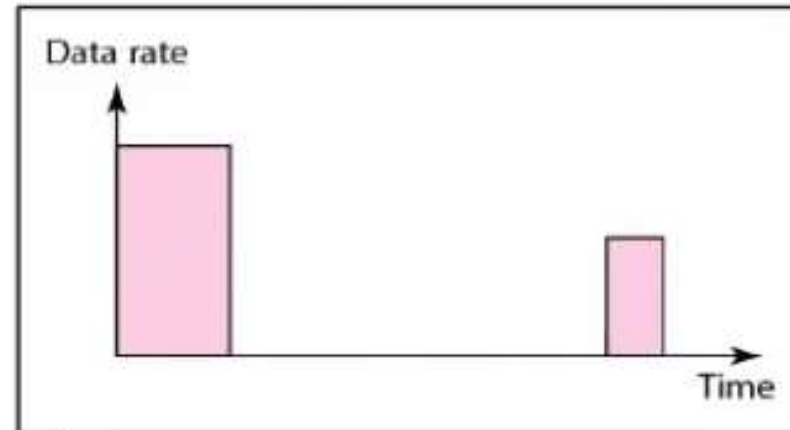
Network Three traffic profiles



a. Constant bit rate



b. Variable bit rate

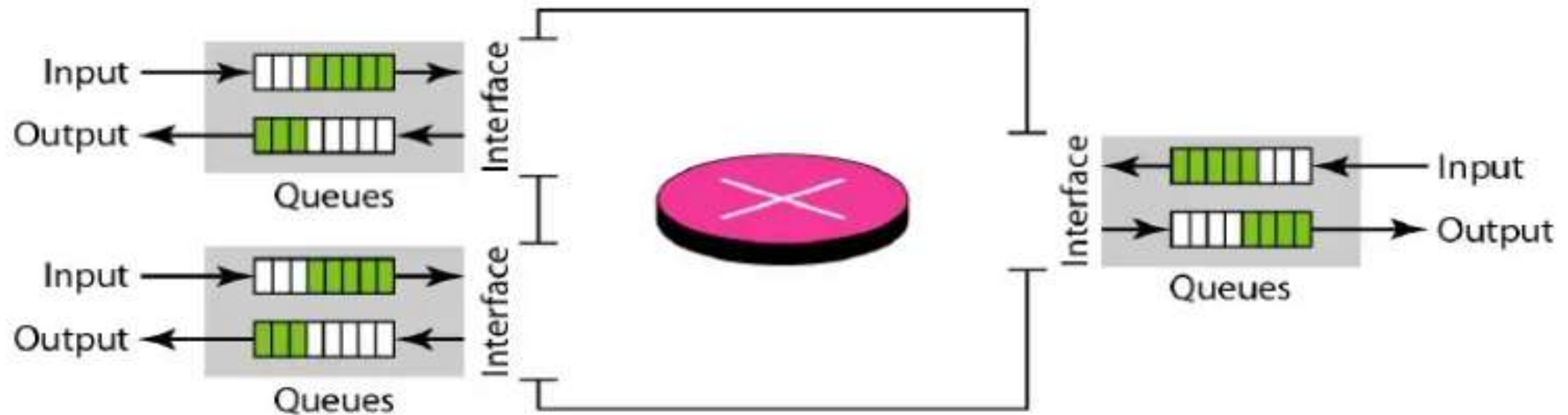


c. Bursty



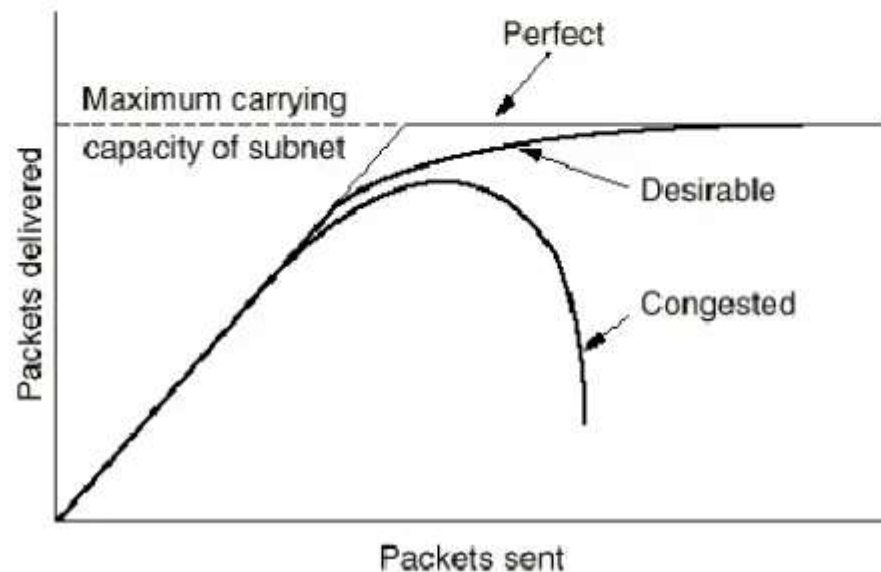
CONGESTION

- Congestion in a network may occur if the load on the **network**—**the number of packets sent to the network**—is greater than the capacity of the network—the number of packets a network can handle.
- Congestion control refers to the mechanisms and techniques to **control the congestion** and keep the load below the capacity.





- ▶ However, the most effective way to control congestion is to reduce the load that the transport layer is placing on the network.
- ▶ This requires the network and transport layers to work together.





Causes of Congestion

- Congestion occurs when a router receives data faster than it can send it
 - Insufficient bandwidth
 - Slow hosts
 - Data simultaneously arriving from multiple lines destined for the same outgoing line.
- The system is not balanced
 - Correcting the problem at one router will probably just move the bottleneck to another router.



Congestion Causes More Congestion

- Incoming messages must be placed in queues
 - The queues have a finite size
 - Overflowing queues will cause packets to be dropped
 - Long queue delays will cause packets to be resent
 - Dropped packets will cause packets to be resent
 - Senders that are trying to transmit to a congested destination also become congested
 - They must continually resend packets that have been dropped or that have timed-out
 - They must continue to hold outgoing/unacknowledged messages in memory.

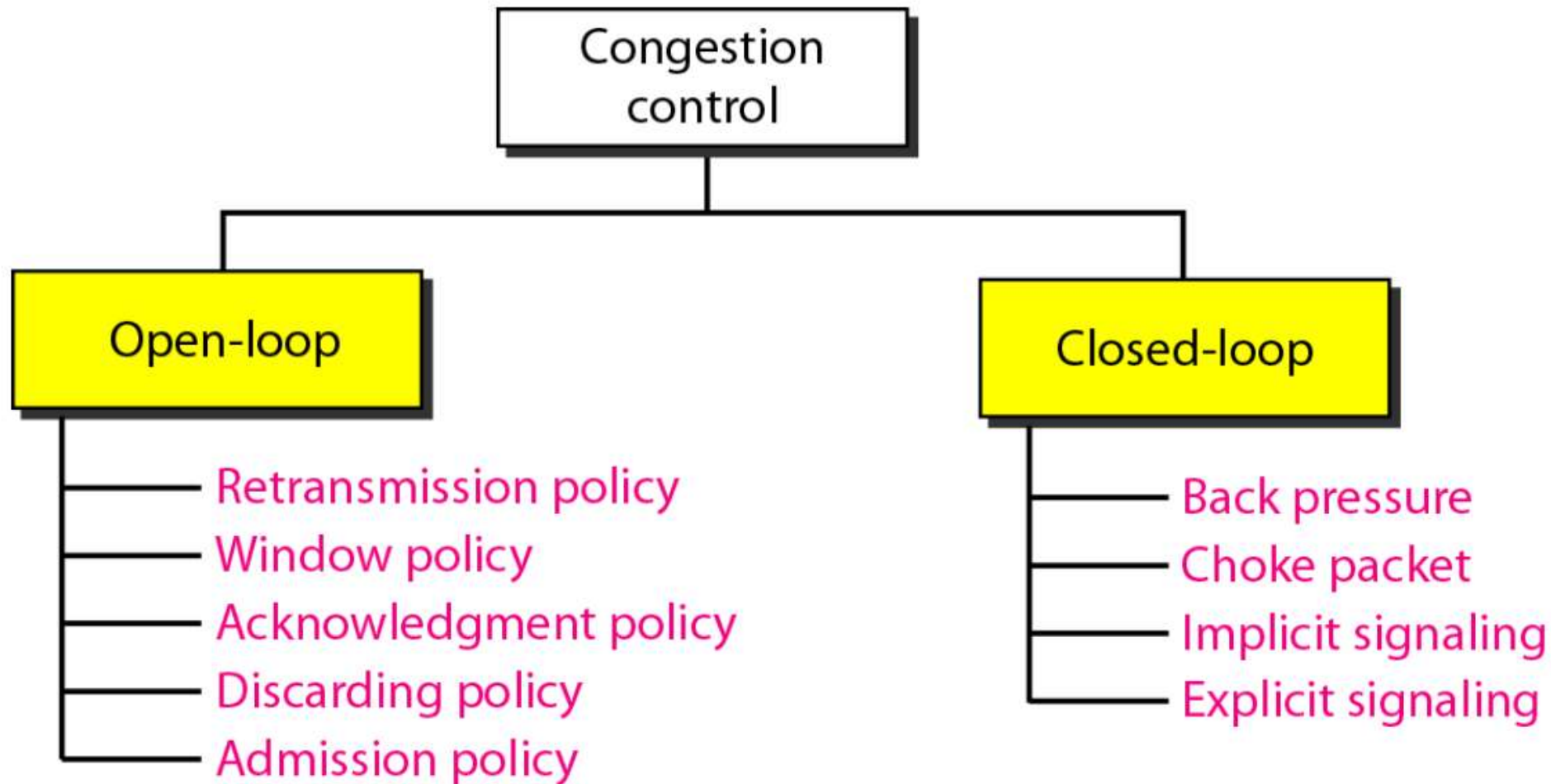


Congestion Control

- Congestion control refers to techniques and mechanisms that can either **prevent** congestion, before it happens, or **remove** congestion, after it has happened.
- In general, we can divide congestion control mechanisms into two broad categories: **open-loop** congestion control (prevention) and **closed-loop** congestion control (removal).
- The open-loop congestion control policies are used to prevent the congestion **before it happens**.
- In open-loop Congestion control is **handled** either by the source or by the destination.
- In closed loop congestion control mechanisms the congestion is **removed after it had happens**.



Congestion control categories

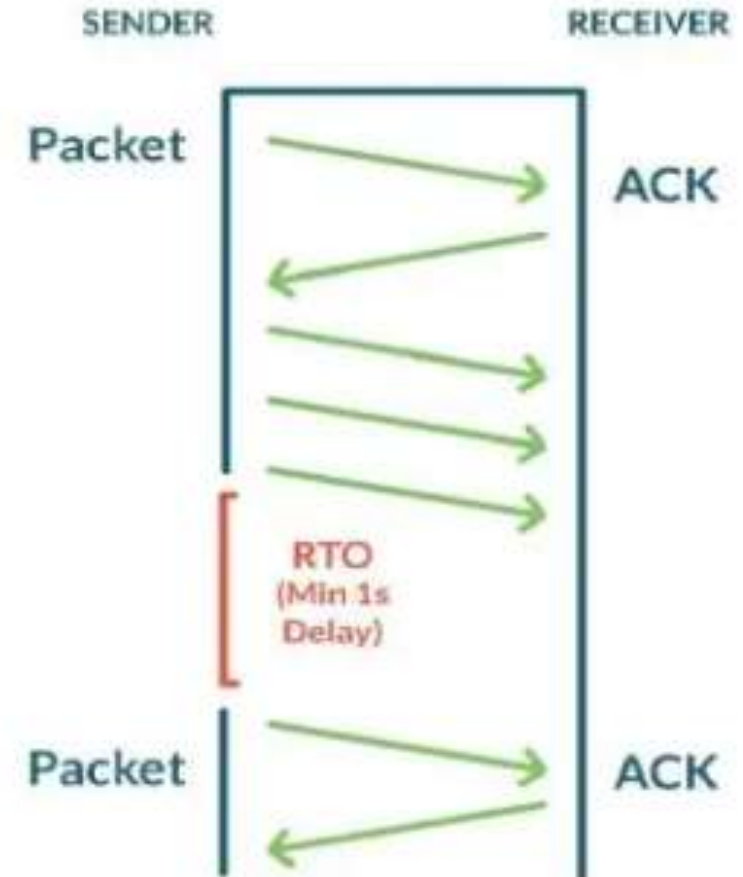




Open-loop congestion control policies

Retransmission Policy

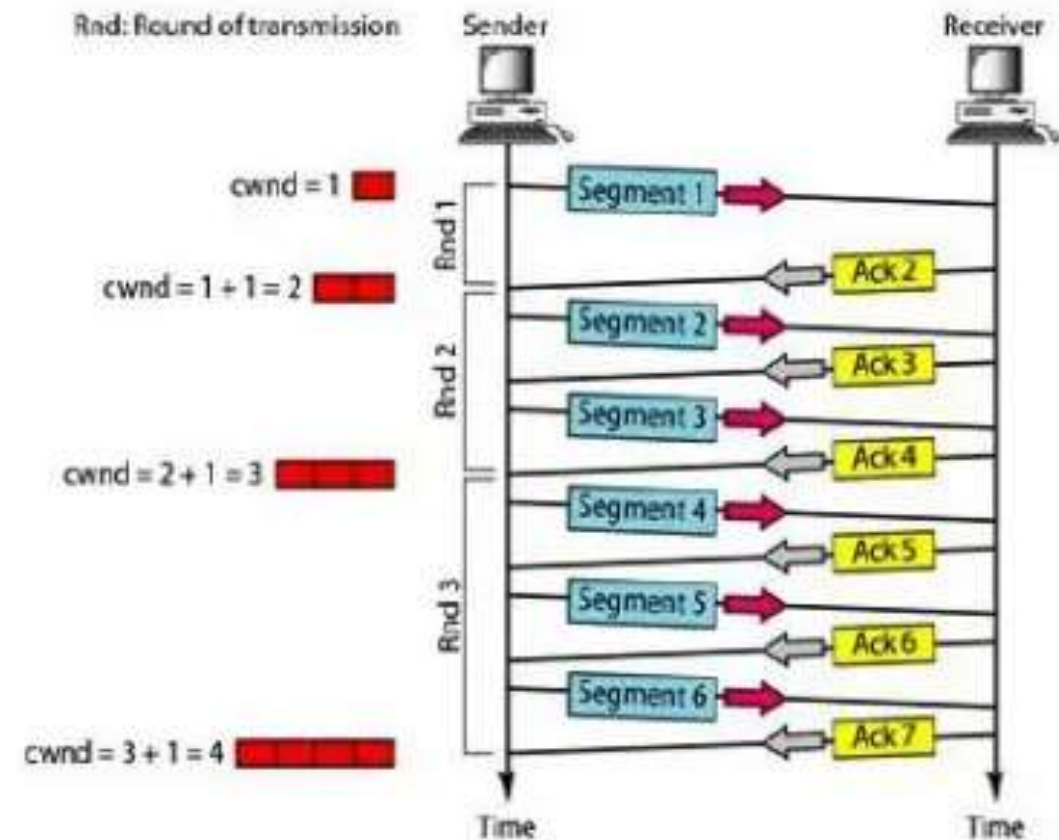
- The sender retransmits a packet, if it feels that the packet it has sent is lost or corrupted.
- However retransmission in **general may increase the congestion** in the network. But we need to implement good retransmission policy to prevent congestion.
- The **retransmission policy and the retransmission timers need to be designed to optimize** efficiency and at the same time prevent the congestion.





Window Policy

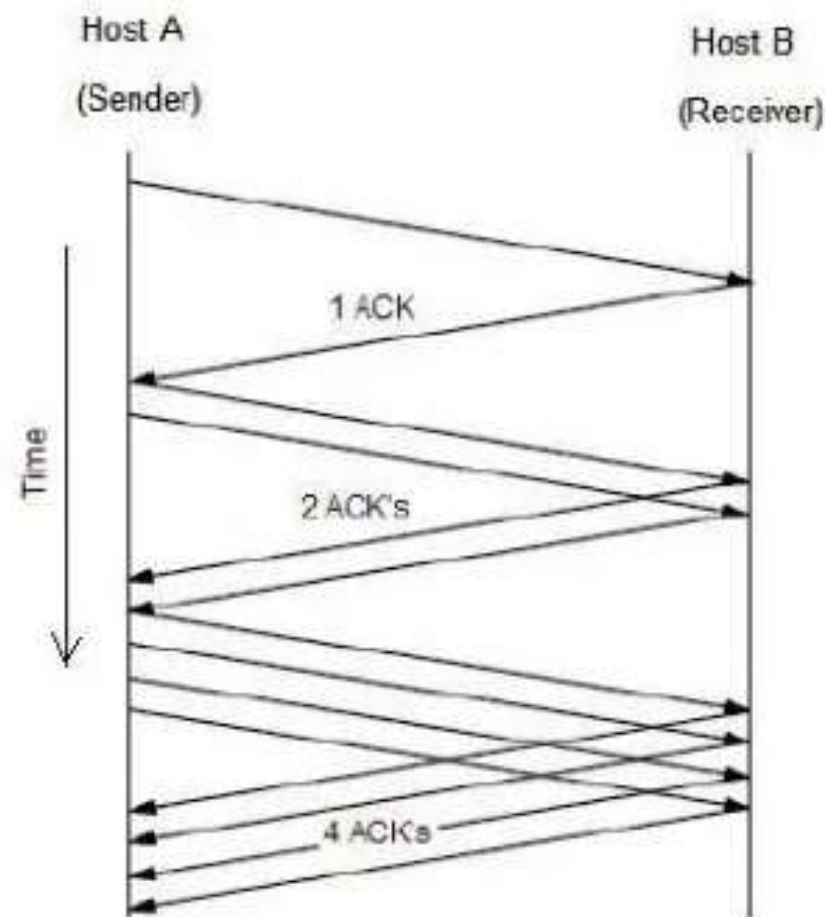
- To implement window policy, **selective reject/repeat window** method is used for congestion control.
- Selective Reject method is preferred over **Go-back-n** window as in Go-back-n method, when timer for a packet times out, **several packets are resent**, although some may have arrived safely at the receiver. Thus, this duplication may make congestion worse.
- Selective reject/repeat method** sends only the specific lost or damaged packets.





Acknowledgement Policy

- The **acknowledgement policy** imposed by the receiver may also **affect congestion**.
- If the receiver does not acknowledge every packet it receives **it may slow down the sender** and help prevent congestion.
- **Acknowledgments** also add to the traffic load on the network. Thus, by sending fewer acknowledgements we can reduce load on the network.
- To implement it, several **approaches** can be used by a receiver:
 - ✓ Acknowledge only if it has a packet to be sent.
 - ✓ May send an acknowledgement when a timer expires.
 - ✓ Acknowledge only **N** packets at a time.





Discarding Policy

- A **router may discard** less sensitive packets when congestion is likely to happen.
- Such a discarding policy may prevent congestion and at the same time may **not harm** the **integrity** of the transmission.

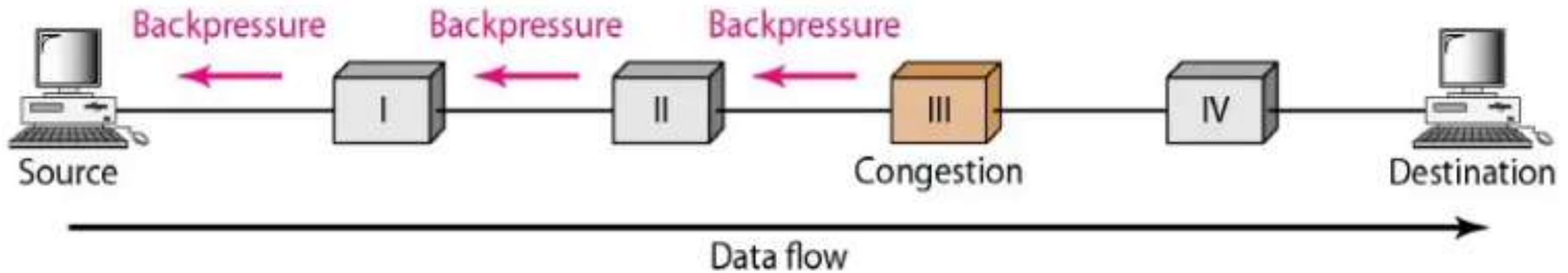
Admission Policy

- An admission policy, which is a quality-of-service mechanism, can also prevent congestion in **virtual circuit networks**.
- Switches in a flow first **check the resource requirement** of a flow before admitting it to the network.
- A router can **deny establishing a virtual circuit** connection if there is congestion in the "network or if there is a possibility of future congestion.



Backpressure

- Backpressure is a **node-to-node congestion** control that starts with a node and propagates, in the opposite direction of data flow.
- The backpressure technique can be applied **only to virtual circuit** networks. In such virtual circuit each node knows the upstream node from which a data flow is coming.
- Here, the **congested node stops receiving** data from the immediate upstream node or nodes. This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream node or nodes.

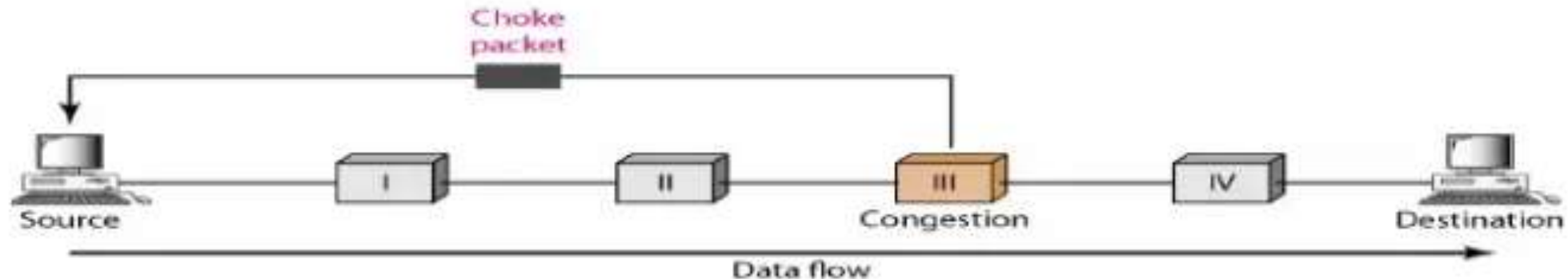


As shown in fig node III is congested and it stops receiving packets and informs its upstream node II to slow down. Node II in turns may be congested and informs node I to slow down. Now node I may create congestion and informs the source node to slow down. In this way the congestion is alleviated. Thus, the pressure on node III is moved backward to the source to remove the congestion.



Choke Packet Method

- In this method of congestion control, congested router or node **sends a special type of packet called choke packet** to the source to inform it about the congestion.
- Here, congested node **does not inform** its upstream node about the congestion as in backpressure method.
- In choke packet method, **congested node sends a warning directly to the source station** i.e. the intermediate nodes through which the packet has travelled are not warned.





Implicit Signaling

- In implicit signaling, there is **no communication** between the congested node or nodes and the source.
- The **source guesses** that there is congestion somewhere in the network when it does not receive any acknowledgment. Therefore the **delay in receiving an acknowledgment** is interpreted as congestion in the network.
- On sensing this congestion, the source **slows down**.
- This type of congestion control policy is **used by TCP**.



Explicit Signalling

- In this method, the congested nodes **explicitly send** a signal to the source or destination to inform about the congestion.
- Explicit signalling is **different** from the choke packet method. In choke packet method, a separate packet is used for this purpose whereas in explicit signalling method, the signal is included in the packets that carry data .
- Explicit signalling can **occur in either** the forward direction or the backward direction
- In backward signalling, **a bit is set** in a packet moving in the direction opposite to the congestion. This bit warns the source about the congestion and informs the source to slow down.

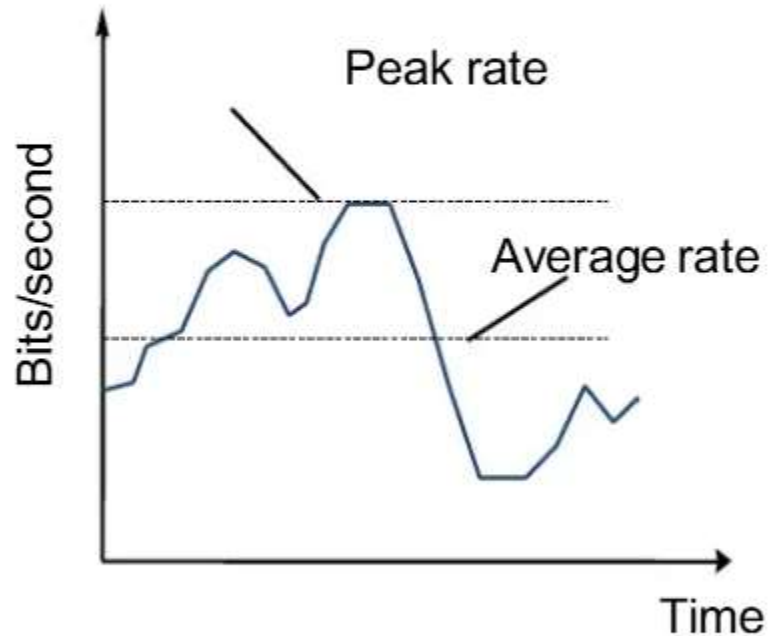


Quality of Service(QoS)

- Network performance is guaranteed to all traffic flows that have been admitted into the network
- Initially for connection-oriented networks
- Key Mechanisms
 - Admission Control
 - Policing
 - Traffic Shaping
 - Load Shedding



Admission Control



Typical bit rate demanded by a variable bit rate information source

- Flows negotiate contract with network
- Specify requirements:
 - Peak, Avg., Min Bit rate
 - Maximum burst size
 - Delay, Loss requirement
- Network computes resources needed
 - “Effective” bandwidth
- If flow accepted, network allocates resources to ensure QoS delivered as long as source conforms to contract



Policing

- Network monitors traffic flows continuously to ensure they meet their traffic contract
- When a packet violates the contract, network can discard or tag the packet giving it lower priority
- If congestion occurs, tagged packets are discarded first
- Leaky Bucket Algorithm is the most commonly used policing mechanism
 - Bucket has specified leak rate for average contracted rate
 - Bucket has specified depth to accommodate variations in arrival rate
 - Arriving packet is *conforming* if it does not result in overflow



Traffic Shaping

- Another method of congestion control is to “shape” the traffic before it enters the network.
- Traffic shaping controls the rate at which packets are sent (not just how many). Used in ATM and Integrated Services networks.
- At connection set-up time, the sender and carrier negotiate a traffic pattern (shape).
- Two traffic shaping algorithms are:
 - Leaky Bucket
 - Token Bucket

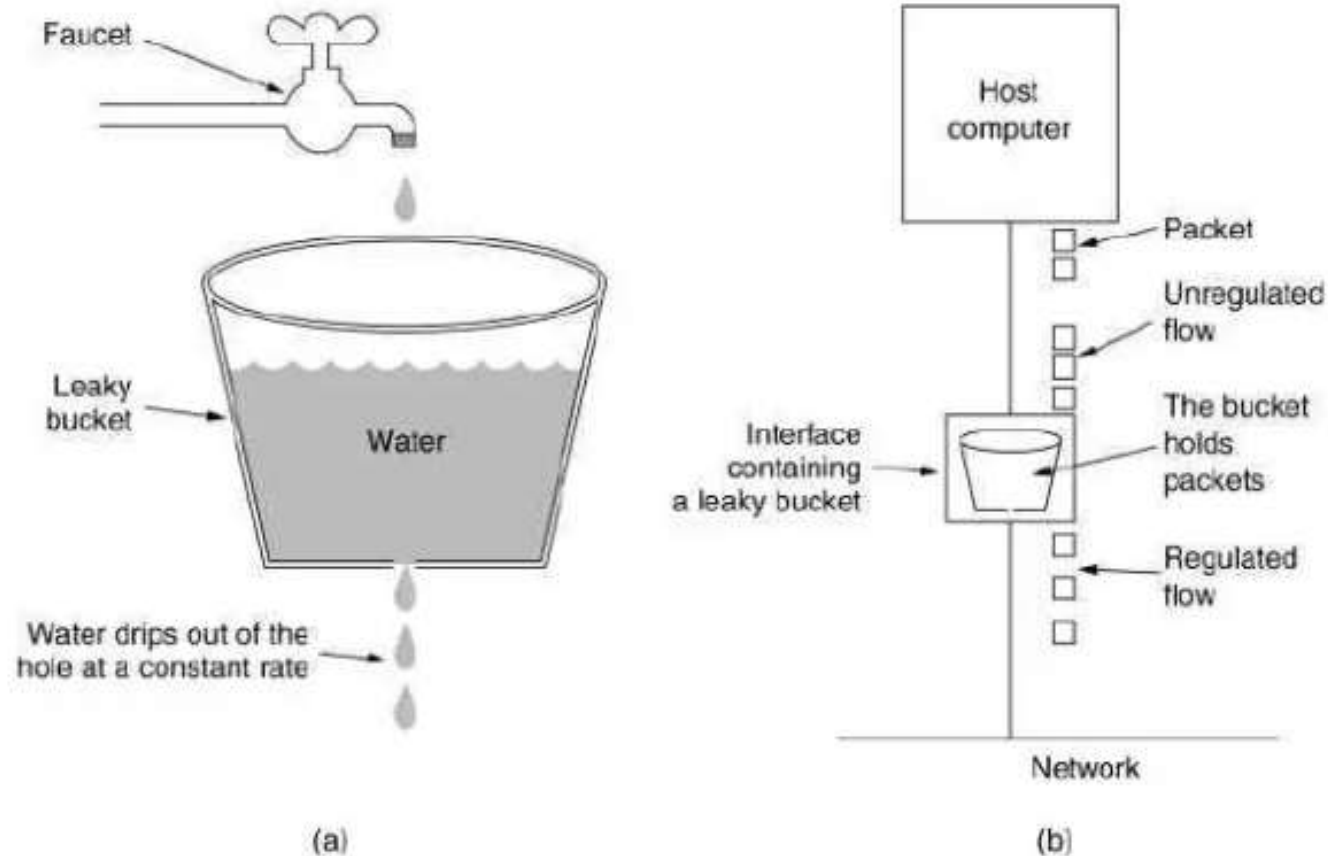


The Leaky Bucket Algorithm

- The **Leaky Bucket Algorithm** used to control rate in a network. It is implemented as a single-server queue with constant service time. If the bucket (buffer) overflows then packets are discarded.
- The leaky bucket enforces a constant output rate (average rate) regardless of the burstiness of the input. Does nothing when input is idle.
- The host injects one packet per clock tick onto the network.
- This results in a uniform flow of packets, smoothing out bursts and reducing congestion.



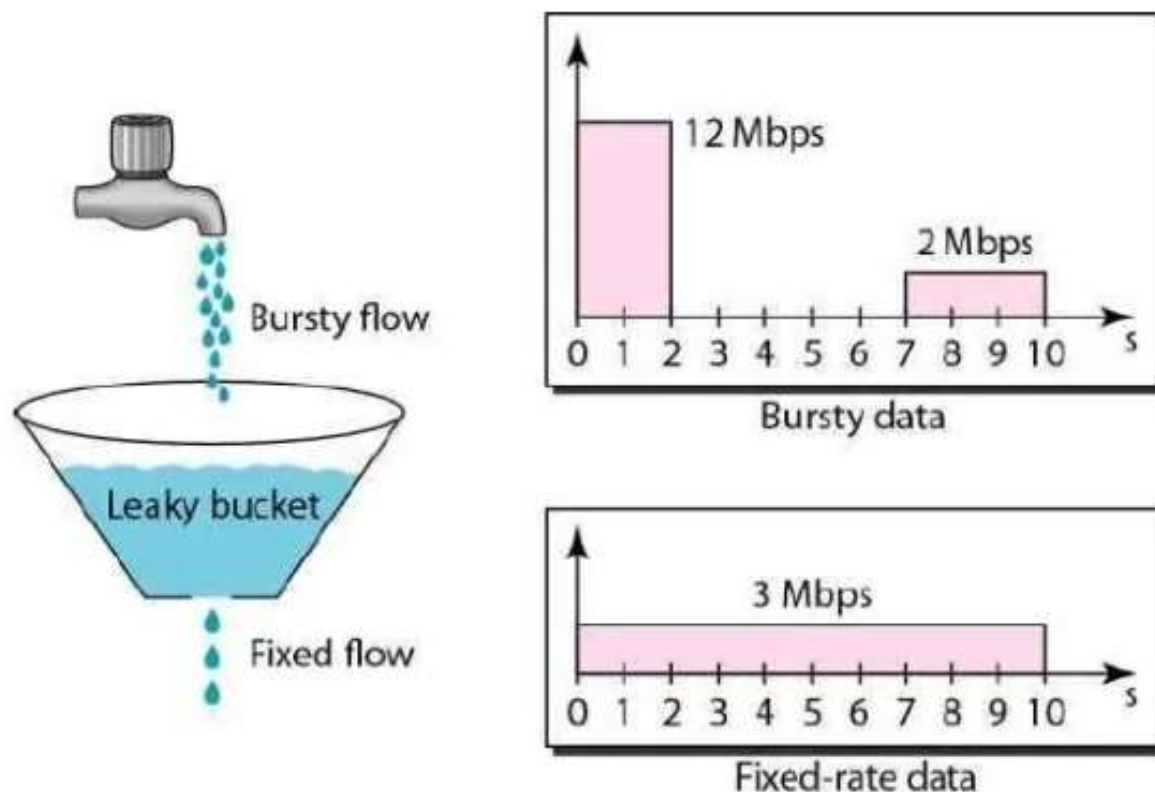
The Leaky Bucket Algorithm



(a) A leaky bucket with water. (b) a leaky bucket with packets.

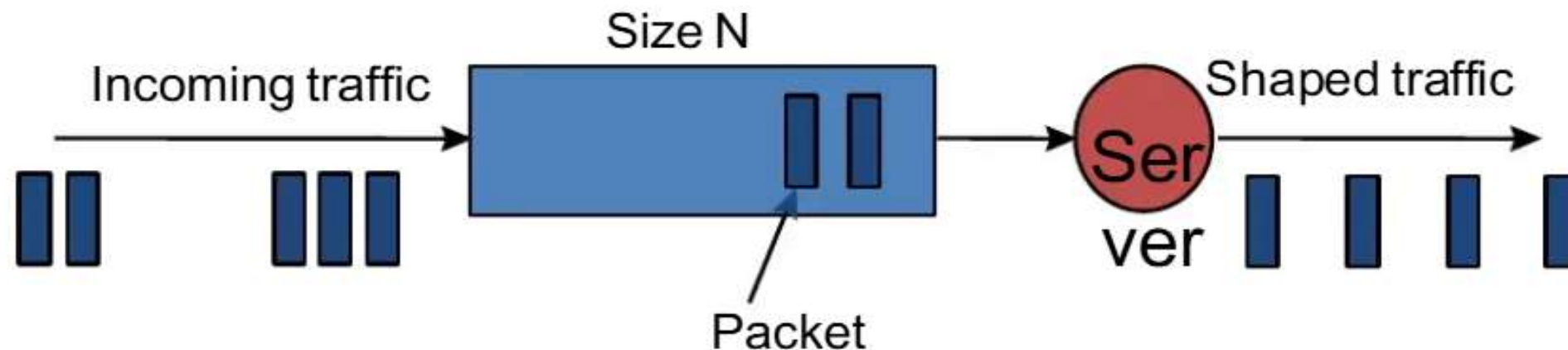


- When packets are the same size (as in ATM cells), the one packet per tick is okay. For variable length packets though, it is better to allow a fixed number of bytes per tick. E.g. 1024 bytes per tick will allow one 1024-byte packet or two 512-byte packets or four 256-byte packets on 1 tick.





Leaky Bucket Traffic Shaper



- Buffer incoming packets
- Play out periodically to conform to parameters
- Surges in arrivals are buffered & smoothed out
- Possible packet loss due to buffer overflow
- Too restrictive, since conforming traffic does not need to be completely smooth



Token Bucket Algorithm

- The leaky bucket algorithm described above, enforces a rigid pattern at the output stream, irrespective of the pattern of the input.
- For many applications it is better to allow the output to speed up somewhat when a larger burst arrives than to lose the data.
- Token Bucket algorithm provides such a solution.
- In this algorithm leaky bucket holds token, generated at regular intervals.
- Main steps of this algorithm can be described as follows: 
 - In regular intervals tokens are thrown into the bucket.
 - The bucket has a maximum capacity.
 - If there is a ready packet, a token is removed from the bucket, and the packet is sent.
 - If there is no token in the bucket, the packet cannot be sent.

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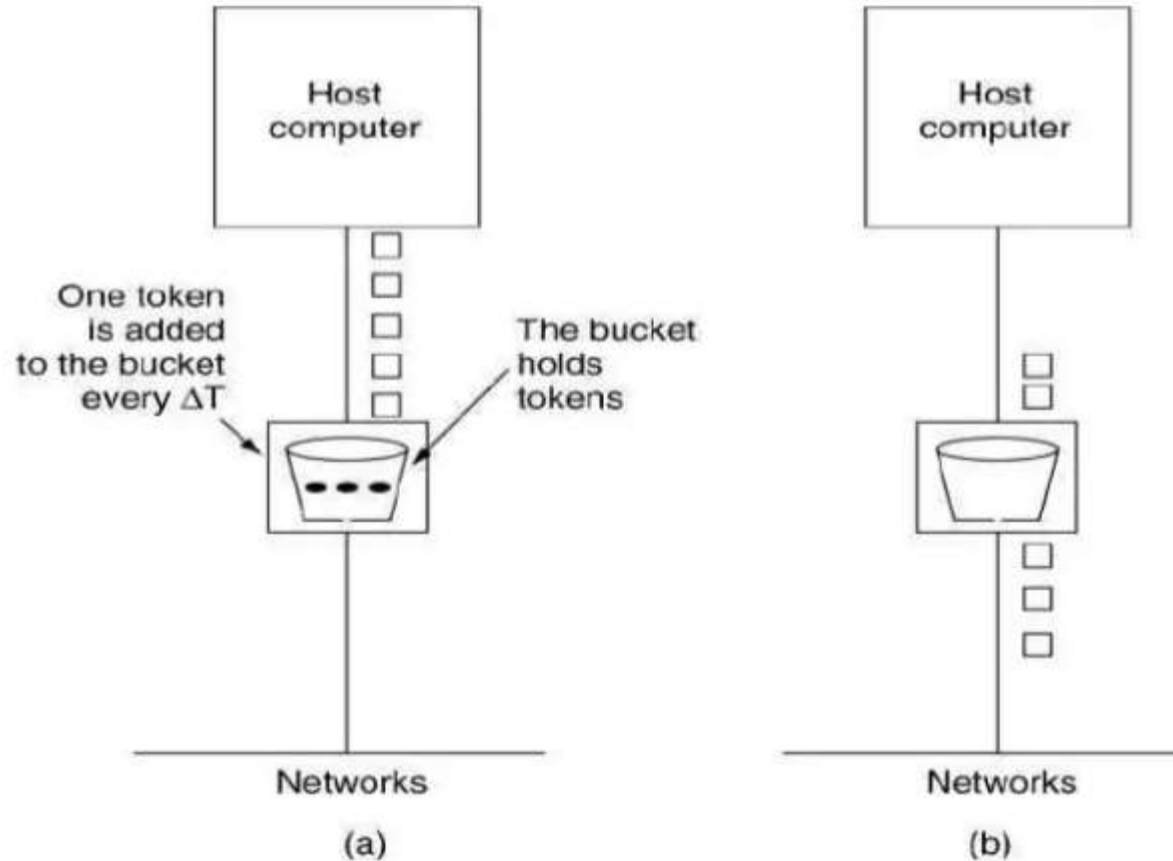


Token Bucket Algorithm

- In contrast to the LB, the Token Bucket Algorithm, allows the output rate to vary, depending on the size of the burst.
- In the TB algorithm, the bucket holds tokens. To transmit a packet, the host must capture and destroy one token.
- Tokens are generated by a clock at the rate of one token every Δt sec.
- Idle hosts can capture and save up tokens (up to the max. size of the bucket) in order to send larger bursts later.

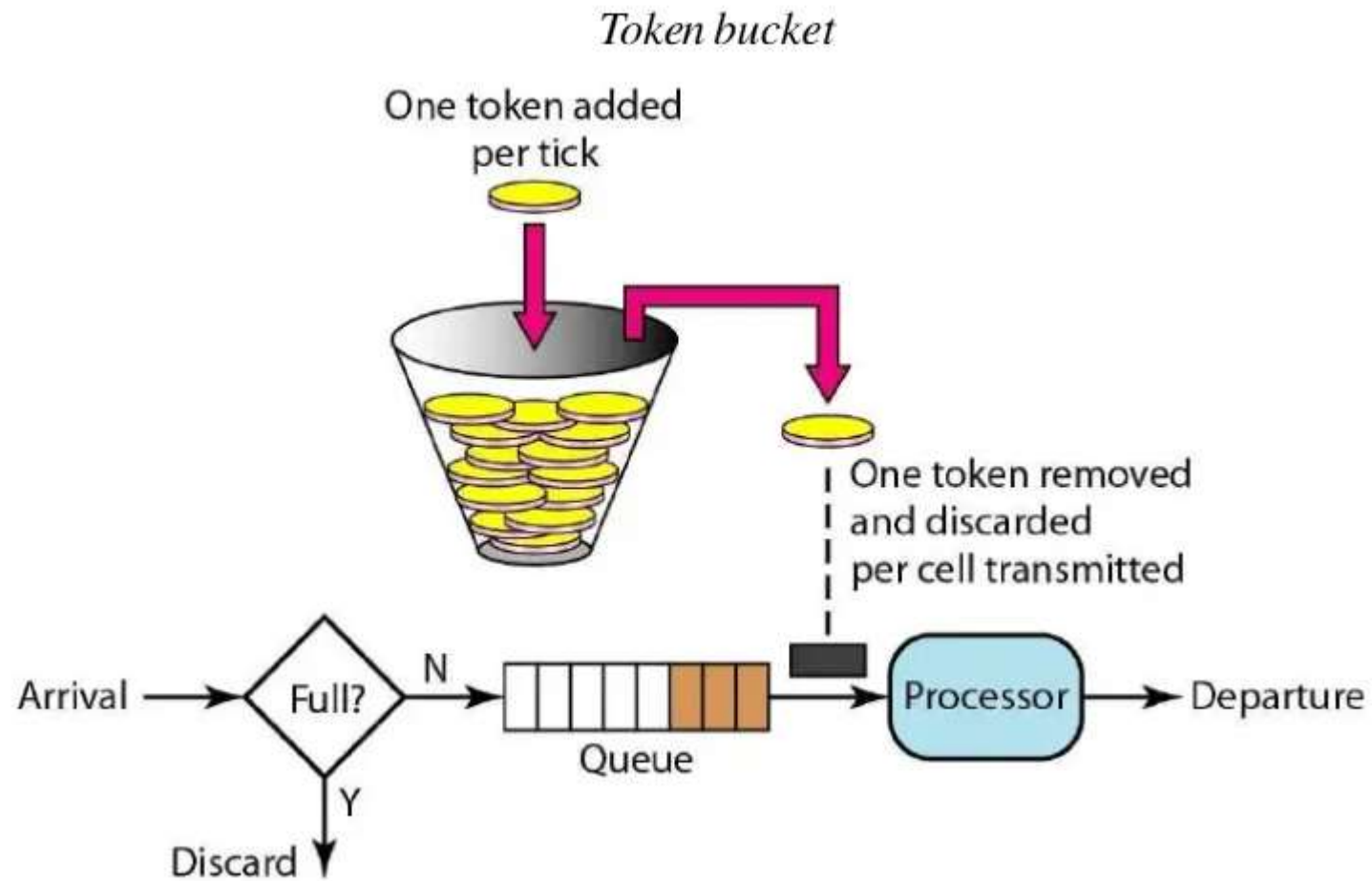


The Token Bucket Algorithm



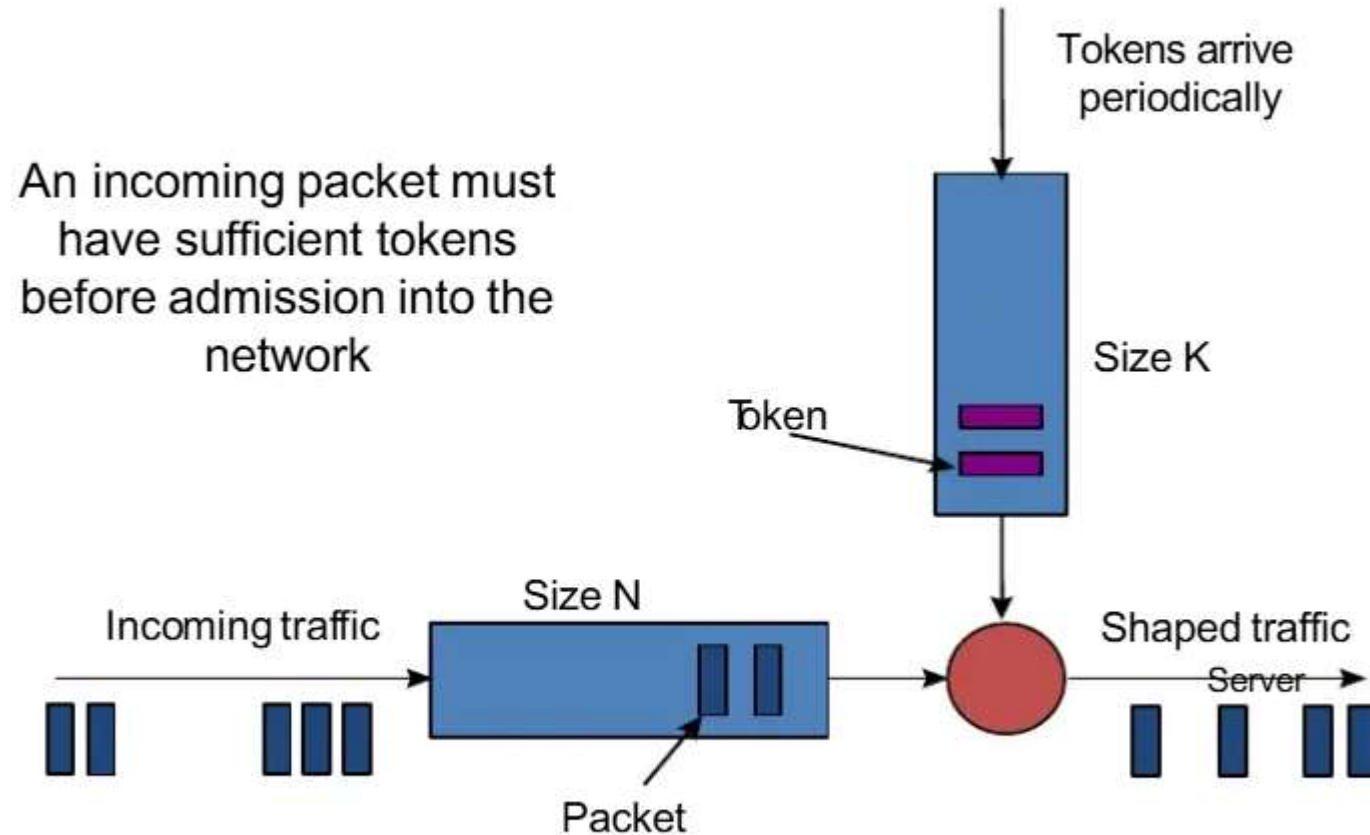
(a) Before.

(b) After.





Token Bucket Traffic Shaper



- Token rate regulates transfer of packets
- If sufficient tokens available, packets enter network without delay
- K determines how much burstiness allowed into the network



Leaky Bucket	Token Bucket
When the host has to send a packet , packet is thrown in bucket.	In this leaky bucket holds tokens generated at regular intervals of time.
Bucket leaks at constant rate	Bucket has maximum capacity.
Bursty traffic is converted into uniform traffic by leaky bucket.	If there is a ready packet , a token is removed from Bucket and packet is send.
In practice bucket is a finite queue outputs at finite rate	If there is a no token in bucket, packet can not be send.



Advantage of token Bucket over leaky bucket

- If bucket is full in token Bucket , tokens are discard not packets. While in leaky bucket, packets are discarded.
- Token Bucket can send Large bursts at a faster rate while leaky bucket always sends packets at constant rate.



Load Shedding

- When buffers become full, routers simply discard packets.
- Which packet is chosen to be the victim depends on the application and on the error strategy used in the data link layer.
- For a file transfer, for, e.g. cannot discard older packets since this will cause a gap in the received data.
- For real-time voice or video it is probably better to throw away old data and keep new packets.
- Get the application to mark packets with discard priority.

